

Venkata karthik Marineni

Data Scientist

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PROFESSIONAL SUMMARY

- Results-driven Data Analyst with 3+ years of experience utilizing SQL, Python, and Excel to uncover trends, optimize reporting, and deliver high-impact business insights that improved operational efficiency, data accuracy, and stakeholder decision-making across diverse domains.
- Proven expertise in supervised and unsupervised learning, feature engineering, and model validation using Scikit-learn, TensorFlow, and XGBoost to enhance customer segmentation, credit risk modeling, and demand forecasting accuracy.
- Strong proficiency in data extraction, transformation, and visualization using tools like SQL, PySpark, Tableau, and Power BI, enabling executive-level decision-making through dynamic dashboards, KPI monitoring, and trend analysis across millions of rows of real-time data.
- Experienced in cloud-native data science workflows using AWS (S3, SageMaker, Lambda), Docker, and Airflow; collaborated cross-functionally with engineers and stakeholders to deploy scalable AI solutions aligned with business objectives, security policies, and regulatory compliance.

TECHNICAL SKILLS

Programming Languages:	Python, R, SQL, SAS
Packages:	Pandas, NumPy, dplyr (R), Matplotlib, Seaborn, Plotly, ggplot2, Scikit-learn, TensorFlow, PyTorch, OpenCV
Data Visualization:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP, Macros)
Machine Learning & AI:	Supervised Learning, Unsupervised Learning, Reinforcement Learning, Neural Networks, Random Forest, Gradient Boosting, XGBoost, SVM, K-Means, KNN, CNN, RNN, LSTM, GAN
Cloud Platforms:	Amazon Web Services (AWS), Google Cloud Platform (GCP)
Databases:	MySQL, PostgreSQL, MongoDB, MS SQL Server
Data Engineering:	Apache Spark, Apache Hadoop, Apache Kafka, ETL/ELT
Other Technical Skills:	SSIS, SSRS, SSAS, Docker, Kubernetes, GitLab CI/CD, Jenkins, Informatica, Talend, MLFlow, Snowflake, Google BigQuery, A/B Testing, Hypothesis Testing, Statistical Modeling, Regression Analysis, Time Series Analysis, KPIs, Accuracy, Precision, Recall, F1-Score, ROC Curve, AUC, Apache Hive, Data Quality & Governance, NLP, Big Data, Advanced Analytics, Data Mining, Data Warehousing, Data Transformation, Data Integrity, Data Cleansing, Data Quality Assurance, Problem-Solving, SDLC, Agile Development, Testing & Deployment
Version Control Tools:	Git, GitHub, SVN
Operating Systems:	Windows, Linux, Mac iOS

PROFESSIONAL EXPERIENCE

Data Scientist | UnitedHealth Group, TX, USA

Aug 2024 - Present

- Developed predictive care management models using **Scikit-learn**, improving disease detection by 24% through classification, feature engineering, and cross-validation, enabling providers to personalize interventions and allocate healthcare resources more effectively.
- Built **LSTM-based time-series forecasting models using PyTorch** to anticipate ICU bed demand and patient readmission patterns, improving hospital resource planning by 21% and reducing patient wait times through sequential modeling and temporal health data analysis.
- Created a **reinforcement learning** triage optimization engine using **OpenAI Gym** and **TensorFlow**, reducing ER overload by 18% through policy-based prioritization of high-risk patients using real-time clinical data and historical treatment patterns.
- Configured **Gradient Boosting models** for clinical outcome prediction, increasing forecast accuracy by 19% and enabling data-driven therapy planning through SHAP-based interpretability, hyperparameter tuning, and early stopping techniques.
- Managed large-scale clinical datasets in **Amazon Redshift data warehouse**, optimizing query performance and ensuring HIPAA-compliant access to real-time analytics dashboards used by clinical and operational leaders for KPI tracking and care quality benchmarking.
- Performed advanced **data mining** on longitudinal EHR datasets to uncover hidden patient risk patterns, increasing early intervention accuracy by 22%, and enabling clinicians to proactively address comorbidities using association rules and clustering techniques.
- Executed an end-to-end **NLP pipeline** for extracting medical entities from clinical notes using spaCy and BERT embeddings, enabling efficient classification of diagnoses, procedures, and symptoms to improve care documentation accuracy and accelerate case review workflows.

Data Scientist | Cognizant, India

May 2021 - Jul 2023

- Engineered scalable data pipelines in **Snowflake**, integrating financial transactions with upstream APIs and market feeds, boosting query performance by 40% using SQL transformations, secure access controls, and real-time synchronization processes.
- Assembled interactive **Power BI dashboards** to visualize revenue trends, loan delinquency, and customer churn, enabling business leaders to take data-driven actions and track KPIs across dynamic financial portfolios with drill-down filters, slicers, and performance segmentation.
- Deployed optimized **XGBoost models** with **Bayesian hyperparameter tuning** to predict loan default risk, increasing model accuracy by 14%, minimizing false positives, and enhancing underwriting workflows using automated validation metrics and threshold calibration.
- Implemented **ARIMA and Prophet time series models** to forecast cash flows and expense trends, improving budgeting accuracy by 23% and enabling strategic treasury planning through seasonality-adjusted projections and robust residual diagnostics.
- Orchestrated robust **ETL workflows using Apache Airflow** to ingest, clean, and transform structured financial data from **AWS S3**, ensuring task dependency resolution, high availability, and seamless delivery to analytics dashboards and machine learning pipelines.
- Applied **Supervised Learning algorithms** to classify high-risk customers, improving fraud detection accuracy and enabling targeted financial strategies using decision trees, logistic regression, and cross-validated ensemble models.
- Designed automated **ETL pipelines using Talend**, integrating data from legacy systems and cloud platforms, enhancing data consistency by 30%, and accelerating financial reporting through transformation logic, error handling, and schema mapping.

EDUCATION

Master of Science in computer science – Purdue University, Fort Wayne, Indiana, USA

Bachelor of Technology in electronics and communication engineering – Vellore Institute of Technology, Vellore, Tamil Nadu, India